

Case Study: Online Course Registration System

Technology Stack

- Frontend: HTML5, CSS3
- Backend: Java Servlets, JSP
- Database Connectivity: JDBC
- Database: MySQL
- Server: Apache Tomcat 10+
- IDE: Eclipse / STS

Case Study Overview

An Online Course Registration System allows students to register, log in securely, browse available courses, and register for courses. This case study focuses on User Registration and Login using Servlets, JSP, JDBC, HTML, and CSS.

Problem Statement

An educational institute wants to digitize its course enrollment process. Students should be able to register, log in, view a dashboard, and log out securely. The system should validate input and store user details in MySQL.

Learning Objectives

- HTML Forms
- CSS Styling
- JSP Pages
- Java Servlets
- JDBC Connectivity
- Session Management
- MVC Architecture
- Form Validation
- Database CRUD Operations

Functional Requirements

Module 1: User Registration

- Full Name
- Email
- Mobile Number
- Password
- Confirm Password
- Validate fields
- Check duplicate email
- Save into MySQL

Module 2: User Login

- Email
- Password
- Validate credentials
- Create session
- Redirect to dashboard

Module 3: Dashboard

- Welcome message
- Display available courses

Module 4: Logout

- Destroy session
- Redirect to login

Non-Functional Requirements

User-friendly interface

Responsive design

Secure password handling (basic)

Proper session management

Input validation

Database Design

Database: course_registration_db

Table: users

id (INT, PK, AUTO_INCREMENT)

fullname VARCHAR(100)

email VARCHAR(100) UNIQUE

mobile VARCHAR(15)

password VARCHAR(100)

SQL Script

```
CREATE DATABASE course_registration_db;
```

```
USE course_registration_db;
```

```
CREATE TABLE users(  
  id INT PRIMARY KEY AUTO_INCREMENT,  
  fullname VARCHAR(100),  
  email VARCHAR(100) UNIQUE,  
  mobile VARCHAR(15),  
  password VARCHAR(100)  
);
```

Project Structure

```
src/  
  dao/  
    DBConnection.java  
    UserDAO.java  
  model/  
    User.java  
  servlet/  
    RegisterServlet.java  
    LoginServlet.java  
    LogoutServlet.java  
WebContent/  
  css/style.css  
  register.jsp  
  login.jsp  
  dashboard.jsp  
  success.jsp  
  error.jsp  
  index.jsp
```

Navigation Flow

index.jsp -> register.jsp -> RegisterServlet -> success.jsp -> login.jsp -> LoginServlet -> dashboard.jsp -> LogoutServlet -> login.jsp

MVC Flow

HTML/JSP -> Servlet -> DAO -> JDBC -> MySQL

Validation Rules

Registration: no empty fields, unique email, matching passwords, 10-digit mobile.

Login: valid email/password, create session.

Session Handling

session.setAttribute("username", fullname);

Display: session.getAttribute("username")

Logout: session.invalidate();

Deliverables

HTML, CSS, JSP, JavaBeans, DBConnection, UserDAO, RegisterServlet, LoginServlet, LogoutServlet, MySQL script, session management, validation, success/error pages.

Expected Outcome

Students will build a complete Java EE MVC web application using Servlets, JSP, JDBC, HTML, and CSS with user registration, authentication, session management, and database connectivity.